Mission 2: MotionLab CubeSat

Mission idea

Students use the CubeSat to measure **motion, acceleration, and orientation** using a **triple-axis accelerometer and tilt/force sensors**. By collecting motion data during different tests, students investigate how forces affect the movement of an object and how engineers design spacecraft to survive motion and impacts during launch, transport, and landing.

Students analyze how **changes in applied force and movement** affect acceleration and stability, helping them understand how spacecraft systems must be designed to operate in dynamic environments.

Driving question

How can a CubeSat measure motion and forces to help engineers understand how spacecraft move and survive launch or impact?

Mission objectives

- Measure acceleration and motion using the triple-axis accelerometer
- Measure CubeSat orientation using tilt sensors
- Investigate how applied forces affect motion and stability
- Analyze how changes in mass affect acceleration
- Evaluate how CubeSat design affects its ability to withstand motion and impact

Example student activities

- Perform controlled drop tests to measure impact forces
- Simulate launch vibrations by shaking the CubeSat
- Measure how acceleration changes when additional mass is added to the CubeSat
- Use the tilt sensor to detect orientation changes and spacecraft attitude
- Graph acceleration data to observe how motion changes over time

NGSS alignment

- **MS-PS2-2:** Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object

- **MS-PS3-1:** Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and the speed of an object
- **MS-ETS1-2:** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem
- **MS-ETS1-3:** Analyze data from tests to determine similarities and differences among several design solutions
- **MS-ETS1-4:** Develop a model to generate data for iterative testing and modification of a design